The Chain of Thought as Compositional Memory in Unmodified Language Models

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Abstract

Causal interventions on six language models show that the chain of thought serves as compositional memory for multi-step reasoning: each written intermediate value is held at its own token position as a register that later computation retrieves through attention, and this stored value, rather than the visible text, determines the final answer. The models are unmodified, used as released with no fine-tuning or task-specific training, and span four families (Qwen, Phi, OLMo, DeepSeek-R1-Distill) on arithmetic and narrative state-tracking tasks. Overwriting the hidden state at one written value's position with the state a different number would have produced, text unchanged, redirects the final answer to that never-written number in 0.70 to 1.00 of cases, while a size-matched random perturbation redirects at most 0.01 on instruction-tuned models. Blocking attention from later positions to that one position collapses the answer's dependence on the value to at most 0.007 on four models, matched blocks of neighboring, random, and earlier-value positions change nothing, and the model continues to produce fluent parseable answers. Layer-resolved sweeps time the retrieval: planting the value through any single layer in the first two thirds of the network succeeds and deeper layers fail, and blocking attention only at the middle third of layers reproduces the full collapse, so the value is read by attention in the middle third and is causally inert afterward. Two positions act as independent stores, and values planted at both compose into their sum, a number appearing nowhere in the input, in 0.88 to 0.90 of cases. Ablating J-space, the concept workspace a fitted Jacobian lens reads from the residual stream, at the strongest dose that leaves ordinary generation intact, does not disturb this externalized computation; this bounds what that readout captures on these tasks rather than placing memory outside the workspace it samples. For oversight, the same interventions show that identical visible text can coexist with different stored values and different downstream behavior, so a chain of thought exposes the symbols a computation operates on without guaranteeing the hidden content behind them.

1 Introduction

When a pretrained language model tracks serial state through a chain of thought, the hidden state at each written value's token is a memory register: settable by intervention, retrieved through attention to that position, and independently addressable, with the stored value rather than the visible text determining the answer. We establish this with causal interventions on six unmodified models from four families (Qwen, Phi, OLMo, DeepSeek-R1-Distill), on two task families with exact ground truth for every intermediate value, and we locate the retrieval in the network's depth.

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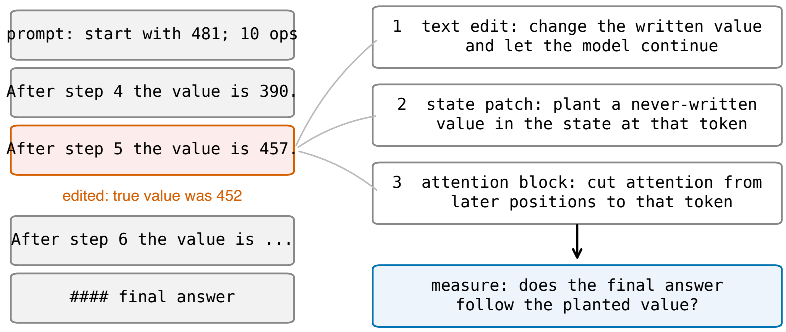

Figure 1: The edit-and-continue protocol. A correct worked solution is truncated after one edited value line and the model continues. Three interventions target the edited value: changing the visible token, overwriting the hidden state at that position with the state a different value would have produced, and blocking attention from later positions to that position. The readout is whether the final answer follows the planted value.

Prior work anticipated pieces of this picture without establishing the mechanism in the wild. Expressivity theory treats generated tokens as the recurrent state of an otherwise fixed-depth computation [Merrill and Sabharwal, 2024], Self-Notes demonstrates writing as working memory behaviorally [Lanchantin et al., 2023], Zhu et al. [2025] show that intermediate-result tokens store mutable values on multiplication and dynamic-programming tasks, and Shih et al. [2026] patch written scratchpad representations on a fine-tuned model. None of these show that unmodified pretrained models do this natively, identify the retrieval route, test whether written stores compose, or map when a model reads a written value rather than recomputing it. These questions are this paper's results:

1. **The hidden state at a written value's position is a value register, natively.** Planting a never-written value into the residual stream at that position, text unchanged, redirects the answer to the planted value at 0.70 to 1.00 across all six models; restoring the correct state under corrupted text reverts the answer at 0.88 to 1.00; size-matched random noise redirects at most 0.01 on instruction-tuned models (Section 3).
2. **Retrieval runs through attention to the written position, before two-thirds depth.** Blocking attention from later positions to that one position collapses the answer's dependence on the value to at most 0.007 on four models in three families, with matched control blocks at the unblocked level and fluency unchanged. Planting through a single layer succeeds at any layer up to two thirds of the network and fails at deeper layers, and blocking attention only at the middle third of layers reproduces the full collapse, so the value is consumed by attention in the middle third (Section 4).
3. **Written positions are independently addressable.** Two never-written values planted at two positions yield their sum, absent from the input, at 0.88 to 0.90; answers using only one of the two occur at 0.01 (Section 3).
4. **A calibrated J-space ablation does not disturb the externalized computation.** Removing the top concepts a fitted Jacobian lens reads from the residual stream, at the strongest dose that preserves ordinary generation, changes neither accuracy nor what gets written. This bounds the particular readout tested; it does not place serial state outside the workspace the lens samples (Section 5).
5. **Reading is the default; recomputation is model-specific.** Models follow an edited written value even when the prompt implies the correct one. DeepSeek V3.2 recomputes only values the prompt generatively defines, and precisely those; Claude Sonnet 4.5 also uses stated checkpoints; Llama 3.3 70B recomputes almost nothing despite solving the same chains from scratch (Section 6).

Table 1: State patch on arithmetic chains at ten steps. Rates are conditioned on items where the text edit was followed (first column), so cross-model comparison of later columns should account for that rate. Brackets are 95 percent bootstrap intervals. The two reasoning-distilled models resolve the problem after their reasoning phase, which raises their random-control reversion; in a logged sample every reverting continuation re-derives from the problem's starting value rather than correcting in place, and re-derivation can only produce the correct answer, so the planted-value column is unaffected.

model	text edit followed	restore correct	plant new value	random control	n
Qwen3-4B	0.98 [0.95, 1.00]	1.00 [1.00, 1.00]	1.00 [0.99, 1.00]	0.00 [0.00, 0.00]	106
Qwen2.5-7B-Instruct	0.85 [0.80, 0.90]	0.97 [0.92, 1.00]	0.74 [0.67, 0.81]	0.00 [0.00, 0.01]	93
Phi-3-medium	0.94 [0.90, 0.98]	0.98 [0.95, 1.00]	0.94 [0.91, 0.97]	0.00 [0.00, 0.00]	102
OLMo-2-7B	0.85 [0.79, 0.91]	0.88 [0.82, 0.94]	0.93 [0.88, 0.97]	0.00 [0.00, 0.00]	100
R1-distill-7B	0.60 [0.53, 0.66]	1.00 [0.99, 1.00]	0.76 [0.68, 0.84]	0.30 [0.22, 0.39]	65
R1-distill-14B	0.44 [0.36, 0.52]	0.99 [0.98, 1.00]	0.70 [0.61, 0.80]	0.21 [0.13, 0.29]	46

2 Tasks and protocol

Both task families give exact ground truth for every intermediate value, so the downstream effect of any edit is computable. Arithmetic chains start from a three-digit number and apply d signed two-digit additions and subtractions. Narrative state-tracking problems describe a character whose count of an item changes through varied events, with varied names, items, verbs, and sentence forms. In both, the model writes one line per step giving the running value.

The protocol takes a correct worked solution, changes the value at one step j, truncates immediately after the edited line, and lets the model continue (Figure 1). A no-edit truncation measures the sampling noise floor, 0.00 throughout. Generation uses temperature 0.6 and top-p 0.95 with five samples per item; unparseable outputs are counted separately. Intervals are 95 percent bootstrap intervals over items; the effects that carry the argument exceed their controls by far larger margins. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-7B-Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; frontier models (DeepSeek V3.2, Claude Sonnet 4.5, Llama 3.3 70B) are tested behaviorally through assistant prefill. Model revisions, prompts, layer bands, and configurations are in the appendix; code and data will be released.

3 The written position holds a settable value register

Table 1 decomposes the effect of editing a written value. Restoring the correct hidden state while the text still shows a corrupted value returns the answer to correct in 0.88 to 1.00 of cases. Planting a third value that appears nowhere in the text redirects the answer to it in 0.70 to 1.00 of cases. The size-matched random control sits at or below 0.01 for the four instruction-tuned models. Because different planted values produce answers following those values, the value content is the causal quantity, and because the planted value is a mid-chain intermediate the model never produced, the patch does not inject the answer. The effect persists with task depth: on Qwen3-4B the planted value is followed at 0.98, 0.97, and 0.95 for chains of 5, 20, and 40 steps. No fine-tuning, task-specific training, or prompt engineering produces this behavior; it is how the pretrained models already work.

The result is a property of written state, not of one template. On the narrative task the pattern replicates: a planted never-written count is followed at 0.83 [0.75, 0.90] on Qwen3-4B and 0.86 [0.79, 0.92] on Phi-3-medium, restoring the correct state returns the answer at 0.89 and 0.94, and the random control is 0.00 in both ($n = 86$ and 89).

Written positions are independently addressable. In a task with two counters summed at the end, planting a never-written value at one final position yields the corresponding mixed sum (0.94 and 0.93 on Qwen2.5-7B, 0.89 and 0.93 on Qwen3-4B). Planting both yields their joint sum at 0.90 and 0.88, at or above the product of the single-position rates (0.89 and 0.83). The random patch leaves the clean answer in place (0.96 and 0.95), and answers using only one of the two planted values occur at 0.01. The model reports a number assembled from two independently set registers.

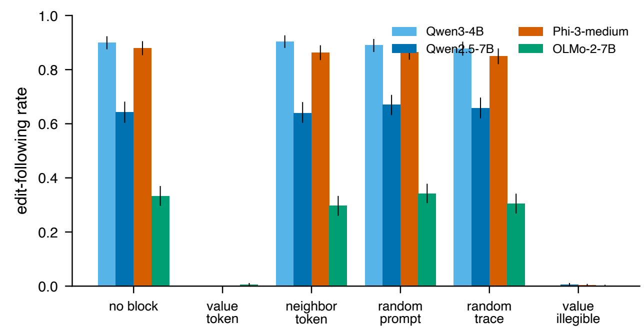

Figure 2: Rate at which the answer follows the edited value under each blocked attention target. Blocking the value's own position collapses following on every model; matched control blocks do not. Error bars are 95 percent bootstrap intervals.

Table 2: Attention block on arithmetic chains: rate at which the answer follows the edited value. OLMo-2-7B continues this format less reliably (low no-block rate) and shows the same contrast.

model	no block	value position	neighbor	random position	earlier value
Qwen3-4B	0.900	0.000	0.903	0.778	—
Qwen2.5-7B-Instruct	0.643	0.000	0.640	0.670	0.658
Phi-3-medium	0.880	0.000	0.863	0.865	0.850
OLMo-2-7B	0.333	0.005	0.297	0.342	0.305

4 Retrieval runs through attention to the written position

Blocking attention from all post-edit positions to the edited value's position collapses the answer's dependence on that value on every model tested (Table 2, Figure 2). Matched blocks of the adjacent words, of a random prompt position, or of an earlier step's written value leave the dependence at its unblocked level. Blocking the operand the next step needs breaks the arithmetic itself, yielding answers that match neither the edit nor the correct value. The collapse holds across three families and on both task formats (natural-language task: 0.730 to 0.007 on Qwen3-4B, 0.720 to 0.003 on Phi-3-medium).

The block removes the value, not the ability to generate. Under the value-position block the unparseable-output rate stays at 0.000 (0.040 on OLMo-2-7B, equal to its unblocked rate). The model produces a fluent, parseable number that matches neither the edited value nor the correct one: it computes forward from a value it can no longer retrieve.

Retrieval occurs in the middle third of the network. Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers). Overwriting the value's hidden state at any single layer from 0 through 22 redirects the answer at 0.90 to 0.94, matching the full multi-layer patch (0.92), while single-layer patches at layer 24 and beyond redirect at 0.02 or less ($n = 60$): the register can be rewritten anywhere upstream of its consumers and nowhere downstream of them. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged (0.857 and 0.863), and no single layer's block matters (0.84 to 0.89 at every layer, $n = 80$): the read is distributed redundantly across the middle third's layers and confined to it. The two sweeps agree on the same boundary, near layer 23: the value is written into the position's residual stream from early depth, retrieved by attention across the middle third, and causally inert afterward. Coarse sweeps on two further models (appendix) show the structure generalizes while the boundary's depth is model-specific: on Phi-3-medium, early- and mid-third patches redirect at 0.89 and 0.87 and late-third patches at 0.007, matching the Qwen3-4B shape, while on Qwen2.5-7B patches in every third redirect at 0.70 to 0.73, so its reads extend into the final third of its 28 layers.

Reading is the preferred path, not the only one. Replacing the value's characters with unreadable dots, without touching attention, also removes following, and differently by model: the Qwen models reconstruct the value from the previous line and recover the correct answer in 0.60 to 0.73 of cases, while Phi-3-medium does not recover (0.00).

5 What a calibrated J-space ablation does and does not disturb

Two results relate the written registers to the model's internal capacity. First, the internal side cannot replace writing on these tasks: instructed to output only the final answer, with token counts confirming nothing else was generated, models fail beyond one to two dependent steps (Qwen2.5-7B falls from 1.00 at $d = 1$ to 0.04 at $d = 4$; DeepSeek V3.2 at 671B reaches 0.23 at $d = 1$ and near zero above), and models write from the first step rather than upon overflow (0.93 to 1.00 of ground-truth intermediates appear in free traces at every difficulty, for every model from 1.5B to 671B).

Second, an ablation of J-space, the concept subspace a fitted Jacobian lens reads from the residual stream [Gurnee et al., 2026], leaves the externalized computation intact. At the strongest dose that preserves ordinary generation (calibrated and frozen on neutral text: next-token agreement 0.87, perplexity ratio 1.12), removing the top 16 active concepts at every position changes neither chain-of-thought accuracy nor bare-answer accuracy nor the amount written, on synthetic chains and on GSM8K, and does no more than a size-matched random-directions ablation. We read this as a boundary on the particular readout tested, for three reasons. The dose is capped at generation-preserving strength, the lens is one fitted artifact, and a written token's contextual representation and its J-space component are not exclusive alternatives, since the lens samples the same residual states our patch overwrites. The J-space account itself reports that explicit chain of thought makes reasoning more robust to J-space ablation and interprets writing as externalizing state the model would otherwise carry internally [Gurnee et al., 2026]. Our result is consistent with that reading and sharpens it with a mechanism: on these tasks the externalized channel is where the serial state causally resides, and the tested lens readout does not capture it.

6 When the written value is read versus recomputed

Models read the register by default. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value the prompt generatively defines (the prompt states the two numbers whose sum becomes the running value) is reverted to correct at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the reversion targets exactly the defined value. Claude Sonnet 4.5 recomputes more broadly, reverting most when a stated checkpoint is the value's only source. Llama 3.3 70B follows edits at 0.97 whether or not the value is prompt-defined, despite solving the same chains from scratch, so the capacity to recompute does not imply recomputing. Qwen models at 4B and 7B never revert. These conditions come from separate experiments with different trace lengths and item sets; each within-experiment contrast is controlled, and we do not claim a single ordered scale across models or that reconstruction cost is the governing rule. Characterizing what governs the choice is open.

The same test bounds the mechanism on natural benchmarks: on GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2, since a benchmark intermediate is often one operation from numbers still in the prompt. Written values act as memory chiefly for state the model does not reconstruct on its own.

7 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which the single-layer sweep in Section 4 addresses. That written intermediates can serve as memory is established: theoretically [Merrill and Sabharwal, 2024], behaviorally [Lanchantin et al., 2023], and causally in specific settings [Zhu et al., 2025, Shih et al., 2026]. We add that the behavior is native to unmodified pretrained models across four families and two formats, the attention-level retrieval route with matched controls, its depth localization, register composition, the qualified J-space boundary, and the read-versus-recompute map. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and filler-token results [Pfau et al., 2024] bound our claims to state the model does not regenerate internally.

8 Limitations

Tasks are numeric and count state tracking, so our claims concern serial task state rather than working memory in general; code, plans, and multi-entity state are untested. The J-space result bounds one fitted lens at generation-preserving doses and does not measure the J-space content of the written positions themselves. The fine layer sweeps are established on one model; the cross-model sweeps are coarse, and on Qwen2.5-7B they do not resolve the read boundary. White-box evidence is at 4B to 14B; frontier evidence is behavioral. The read-versus-recompute results support within-experiment contrasts, and the only patch control is size-matched noise; a structured control planting a valid activation for a different quantity would further isolate value content.

9 Conclusion

On serial state-tracking tasks, unmodified pretrained language models use the hidden states at written intermediate-value positions as compositional registers: settable by intervention, retrieved through attention in the middle layers, independently addressable, and decisive for the answer, while a calibrated J-space ablation leaves the computation undisturbed. For oversight the implication cuts both ways: the written positions are a causally used memory channel, and identical text can carry different stored values, so trace monitoring observes the channel while verifying the state behind it remains a separate problem.

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